

CORRADO LOCATI



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🌐 Portfolio: corradolocati.github.io/corrado-portfolio | 🐙 GitHub: github.com/CorradoLocati

Italian scientist with extensive expertise in electrochemistry, materials science, and data analysis. Dedicated team leader, able to manage time and tasks for the group, internal conflicts, and to successfully report tasks outcomes to internal and external stakeholders.

Fast learner, excelling as an independent worker, as well as being part of international teams. Known for structured approaches, critical thinking, strong work ethics, and enthusiasm for new challenges.

WORK EXPERIENCE

Data Analyst | London School of Economics and Political Science (Online) | 08/07/2024 – 24/02/2025

- Conducted data analysis for a financial data solutions provider using Python. Delivered a turnkey solution for buying and selling stocks based on time prediction.

Senior Electrochemist | Elestor B.V. (Arnhem, NL) | 01/01/2023 – 01/09/2024

- Led a team of five in researching novel platinum-based catalysts for bromine redox flow batteries; achieved an 80% improvement in the company's knowledge of catalysts (e.g., ECSA calculation).
- Coordinated catalyst synthesis and characterization with internal and external stakeholders. Developed one novel Pt-based catalyst and two synthesis techniques; achieved 100% improvement in company knowledge on catalyst resistance to HBr solutions.
- Conducted electrochemical tests (CV, EIS, charge-discharge) on a three-electrode setup as well as on test stands (2 and three electrodes, with 50 μm Pt wire as pseudo reference electrode). Implemented a turnkey solution to test cells.
- Contributed to discussions and data analysis on corrosion and cell performance with other company departments. Delivered 50% improvement on corrosion analysis.
- Authored more than ten SOPs and other documents on procedures and tests.

Application Scientist | Metrohm Autolab (Utrecht, NL) | 01/11/2012 – 01/11/2022

- Performed laboratory experiments to write application notes covering different electrochemical domains, like corrosion (following ASTM, NACE, and ISO standards), battery testing (EIS, charge-discharge, GITT, PITT), solar cells (I-V tests, IMVS, IMPS), and electrocatalysis (EIS, CV with ECSA calculations). Delivered more than 40 application notes and NOVA and INTELLO electrochemical procedures.
- Delivered more than 20 SOPs and more than 10 QA/QC documents in compliance with ISO 9001.
- Actively contributed to the ASTM G01.11 committee. Delivered a collaboration for developing a standard about in-situ EIS measurements.
- Managed product development; launched new products including the battery app and duo coin-cell holder.
- Managed laboratory and demo warehouse; improved lab efficiency by 80% and warehouse utilization by 60%.
- Delivered lectures and workshops to internal and external stakeholders across Europe, Asia, Australia, and North America on electrochemistry, EIS, and instrumentation. Trained more than 100 area managers and students.

Postdoctoral Researcher | TU Delft (Delft, NL) | 01/01/2012 – 01/06/2012

- Electrospray technology for drug delivery.

Postdoctoral Researcher | TU Delft, Kema, ARN, RUPS (Delft, NL) | 01/01/2011 – 01/01/2012

- Development of a membrane for Li-ion recovery from battery waste.

Postdoctoral Researcher | TU Delft, DSM (Delft, NL) | 01/01/2010 – 01/07/2010

- COMSOL calculations on thermal and electrical conductivities in polymer blends for materials used in LED lighting.

EDUCATION

Career Accelerator in Data Analysis | London School of Economics and Political Science (Online), | 08/07/2024 – 24/02/2025

Comprehensive training in data wrangling, statistical analysis, predictive analysis, A/B testing, decision tree analysis, wordcloud, using Excel, Python, R, SQL, and Tableau. Hands-on collaboration with industry partners.

PhD in Chemical Engineering | TU Delft (Delft, NL) | 01/01/2006 – 03/02/2012

Thesis: Interaction between High-Voltage Cathode Materials and Ionic Liquids for Novel Li-Ion Batteries.

Focus: Synthesis, structural and electrochemical characterization of high-voltage nanostructured cathode nanomaterials. Study the interface between cathode material, commercial electrolytes, and ionic liquids. Building of coin cells and pouch cell batteries. Battery test with CV, charge/discharge, and EIS. Combining electrochemical techniques with EXAFS and XANES.

Second-Level Master in Material Science for Micro and Nanotechnologies | IUSS (Pavia, IT) | 01/02/2005 – 01/02/2006

Synthesis, structural and electrochemical characterization of high-voltage nanostructured cathode nanomaterials.

Master's Degree in Material Science | Milano-Bicocca University (Milan, IT) | 01/10/1995 – 09/03/2004

Theoretical and practical knowledge in semiconductors, polymers, synthesis, and characterization of materials.

Thesis: Interaction of Boron Trifluoride with Defective Sites of Binary Oxides (theoretical chemistry).

SKILLS

Technical Skills

- Material science. Electrochemical Techniques (e.g., CV, EIS, GITT, PITT, LP, PDP). Structural characterization of materials (XRD, XAS). Thermodynamic techniques (DSC, TGA), Gaussian98.
- Product development & Management, Project Management Tools (Miro, Asana).
- Python, SQL, R, Octave, Tableau, Microsoft Office (Word, Excel, PowerPoint), machine learning and artificial Intelligence.

Core Competencies

- Teamwork, Problem Solving, Communication, Adaptability, Planning, Self-Management, Observation, Decision Making, Collaboration, Diplomacy, Empathy, Professionalism, Ethical Responsibility.

CERTIFICATIONS AND TRAININGS

- AI in Data Analytics – FourthRev Learning Track – FourthRev, London, UK (18/07/2025)
- Video Production for Marketing & Training – Metrohm Autolab, Utrecht, NL (12/2021)
- Product Management Bootcamp – Tarigo, UK (03/2021)
- Voltammetry Basics – Metrohm AG, Herisau, CH (02/2018)
- Teaching Course – Metrohm AG, Herisau, CH (07/2016)
- Business Presentation Skills – Van Kessel Trainings, Loosdrecht, NL (01/2015)
- Spectroelectrochemistry – IFW Leibniz Dresden, De (08/2013)
- Electrochemistry Winter School – University of Bath, UK (01/2013)
- Scientific Writing in English PROM4 – TU Delft, NL (05/2009)

SCIENTIFIC PUBLICATIONS

- Nanostructured Fe₂O₃ and CuO Composite Electrodes for Li-ion Batteries, J. Power Sources, 2011,
- Nanosized High-Voltage Cathode Material: Structural, Electrochemical and In-Situ Investigation, J. Power Sources, 2009
- Interaction Between Carbon Dioxide and Ionic Liquids: Novel Electrolytes for Li-Ion Batteries, J. Power Sources, 2009
- Mg-Doped LiNi_{0.5}Mn_{1.5}O₄ Spinel for Cathode Materials, J. Power Sources, 2007
- Nanopowders of Spinel-Type Electrode Materials for Li-Ion Batteries, SSI, 2006
- Probing the Basicity of Regular and Defect Sites of Alkaline Earth Metal Oxide Surfaces, ChemPhysChem, 2004